

Parking Booking System: Reducing Congestion Through Innovation

Classes: LD11

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Programming For Business

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Final Project

Programming for business

Book Parking System: Reducing Congestion Through Innovation

I. Project Background

Parking space has been a continuing issue in big cities worldwide (Farda & Balijepalli, 2018). It leads to a multitude of troubles, including congestion–both in the parking space and the streets leading to it, delays in activities, and increased emissions. This issue is a factor in Jakarta being in the top 10 worst cities in the world in terms of street congestion (tricruise.id). And not just that, Jakarta citizens spend on average 22 days a year amidst congested traffic (Jakartapost).

Every year, as the student count of BINUS University grows, the amount of vehicles used grows alongside it. And with it, the issue of limited parking spaces and traffic congestion becomes bigger every year. The congestion issue in-and-around the parking space during peak hours can lead to unsafe driving from people going through the streets, and cause near-misses and accidents more frequently (Abalate & Fageda, 2021).

As students, we ourselves have experienced the congestion in real time during our commute to and from campus. From exiting the parking space, all through to the main road. And it is worsened by the search for parking space, which sometimes need multiple roundtrips to even find one, or even not finding a space at all. This not only wastes time that could be used for attending classes and other academic activities, it also creates frustration that can lead to lowered productivity potential in class (Simatupang & Tantri, 2024).

One way to combat the issues stated above is to use an IoT-based system to check parking space availability and an AI-based prediction algorithm to predict trends and spikes in usage throughout the day to find a parking spot, even on a busy day. This technology can reduce time loss in finding a parking space, this reducing emission as a byproduct (Puri & Chaudari, 2025). Moreover, our solution also has a booking system that allows students and employees to book a parking spot up to 48 hours in advance, ensuring minimal time loss and a cleaner parking experience.

II. Project Problem

- **Limited Parking Space**

  With the number of students, lecturers, and staff increasing every year while the parking capacity remains limited, it has become difficult to find available parking spaces, especially between working hours (09:00 and 17:00).

- **Traffic Congestion**

  The difficulty in finding parking spots leads to congestion within the parking area and on the surrounding roads near the campus. This occurs because students who cannot find proper parking spaces are forced to park their vehicles outside designated areas or to go around the parking space multiple times to scout out potential or empty spaces.

- **Wasted Time Searching for Parking**

  Many users have to circle around repeatedly to find an empty spot, which results in delays in attending classes or other academic activities among other problems.

III. Methodology

1. Data Collection & Analysis

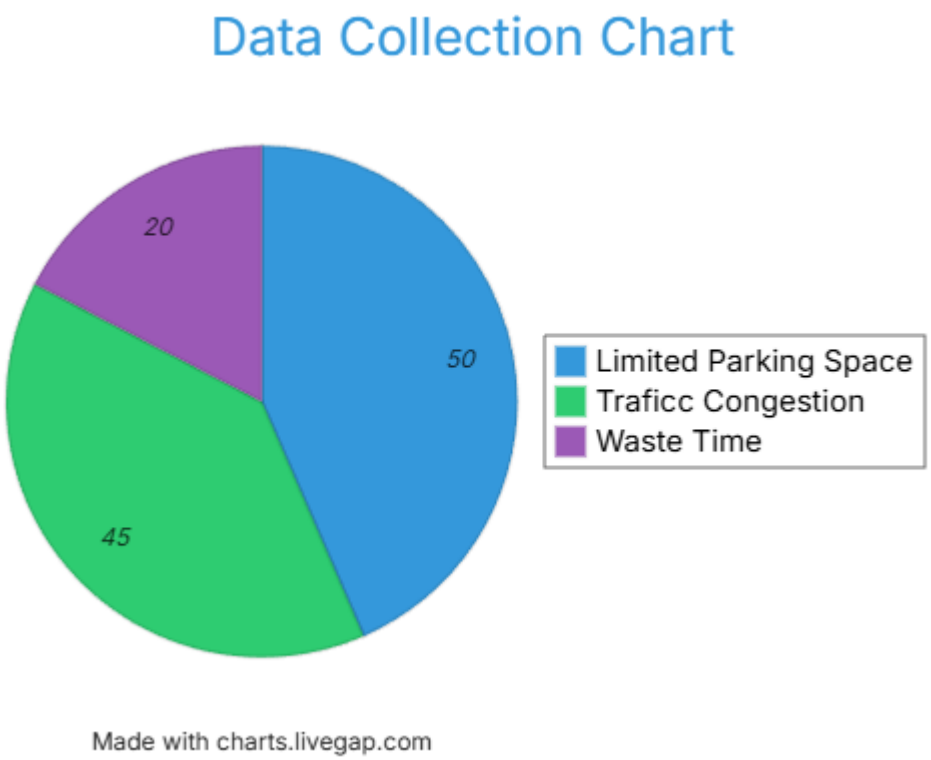
Before entering the design phase, we validated the problem through:

- Field Obeservation:
  Conducted in the campus parking area during peak hours (09:00 – 17:00).

Objective: Measure the average time wasted when looking for parking ((multiple roundtrips) and map student parking patterns, such as areas near the exit that fill up faster during morning classes.

- User Interview :
  Carried out to students and staff

  Objective: Validate user frustration levels and gather feedback on the most needed features, such as booking features and delay time tolerance).



The following are the results of data collection that we have grouped based on the questionnaire we have made. Here it can be explained that of the 50% of the data, most of it occurs due to limited parking spaces, then 45% also occurs due to congestion in the Jakarta area, namely in the binus area itself. Of the 20% who waste time, there are still people who are confused when looking for parking because they waste time in the parking lot.

2. Kebutuhan Pengguna

Reservations	Navigation	Data Real Time	Verification & Penalties
The uncertainty of finding parking is a major source of stress. Therefore, the system should allow users to secure parking slots in	To eliminate "recurring rounds" the system should allow users to secure parking slots in advance, by prioritizing those with verified onsite schedules.	The system should provide live status updates, such as filled/empty with minimal latency to prevent users	To ensure that the booking system is not misused, the system requires a strict mechanism (i.e. QR Code scanning and AI verification) to verify the physical presence of the

advance, prioritizing those with verified onsite schedules.		from driving to a newly filled spot.	vehicle and imposes fines (penalties) on users who order but are absent or late to confirm.

3. System Flow

A. Infrastructure Explanation

1. LED Indicator Light (Visual Guide)

In each parking slot, one indicator LED light is installed on the palfon. These lights connect to sensors and databases to provide direct clues to anyone on site:

GREEN : Empty spot and free to be occupied by anyone

BLUE : Empty spots but have been booked through the user application without booking are prohibited from parking.

RED: The spot is being filled by vehicles.

2. Concept: Hybrid Parking"

- Booking User: Users who want certainty of place. They order through the app, get spots with blue lights, and are verified by AI Cameras and QR.
- Old Users: Old Way users who come immediately, they can still park in the green light spot. If they are desperate to park at a blue light, the system will detect the violation.

B. System Workflow

1. Reservations
   - Users open the application, sistem memvalidasi jika ada jadwal kuliah maka user diizinkan memesan.
   - The user chooses a parking spot (misal F2-53) and enters the data of the vehicle to be parked.
   - The data is stored and the user gets a digital QR Code. The LED lights in the parking spot change from green to blue.
2. Arrival & Detection AI Camera (LPR)
   - When the vehicle enters the F2-53 spot, the AI Camera automatically captures the license plate number.
   - The system matches the AI-captured license plate with the data that has been booked

- If the plate number is different from the one registered at the time of booking, then the system sends a warning notification to the user's cellphone.

3. Attendance Confirmation
   - The user got out of the vehicle and showed the QR Code on his cellphone to the scanner machine on the parking pole.
   - The system combines Valid QR data and Vehicle Numbers captured by AI Cameras
   - If the data matches then the LED light changes to Red. and booking status is Confirmed
4. Finished Parking
   - Ultrasonic sensor detects the car out
   - The system changes the LED Light back to Green. The spot is now available again to other users.

5. System Design

In the system design we created here, we created a Use Case Diagram that presents the functional interaction between users and admins in a parking system that we designed to solve parking congestion problems through digital reservations and AI-based automation. The diagram we created became the conceptual foundation of how the system works from the perspective of users and managers

A. Identification of actors

There are 2 main actors with different but complementary responsibilities:

User

Users are the main actors who use parking services. In the context of implementation, users can be students, staff and even visitors who have access to the campus parking system.

Admin

The admin here acts as an actor who manages the system who is responsible for data validation, operational configuration, and evaluation of the overall performance of the parking system.

B.Use case By User

1. Register

This use case is the initial stage before the user can use the system. The user registers the identity and vehicle data into the system.

This data is the basis for AI Camera (License Plate Recognition) to recognize vehicles when entering the parking area without the need for manual checks

2.Book a parking spot

After logging in, the user will make a reservation for a parking spot before going to the location. The system will check spot availability based on the prevailing rules and rules,

This use case is at the core of the system because it plays a direct role in reducing Queues and parking searches are the main causes of congestion.

3. View Parking Spot Status

Users can see the status of parking spot availability in real-time. In the concept of Hybrid Parking, this feature allows:

Users with bookings (reserved)

Users without booking (go-show)

to get visual information related to spot availability (e.g. through LED color indicators).

4. Receive Navigation Guide

After a successful booking, the system provides navigation guidance to the designated parking spot. This feature minimizes unnecessary vehicle movement within the parking area, thereby improving internal traffic efficiency.

5. Accepting Barcodes

Users receive a QR Code as proof of digital reservation. This QR Code is used for:

- Login validation
- Synchronization with scanner system and AI Camera

6. Paying Penalty Fines
   When the user makes a booking and is not detected by the user parking the vehicle at the booked spot, they will get a penalty in the form of cash.

C.Use cases by Admin Actors

1. Managing User Data and Lecture Schedules

Admins manage user data, vehicles, and schedules (e.g. lecture schedules or operating hours). This data is used by the system to:

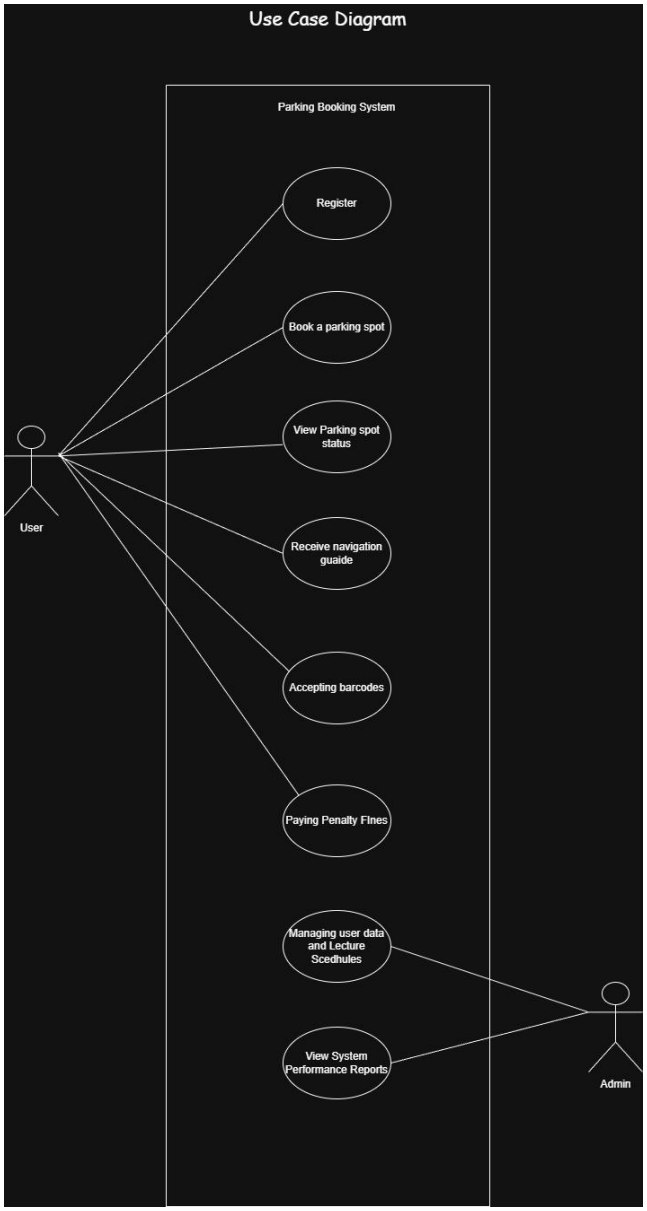
- Determine booking priority
- Limit reservations to specific times
- Becoming an AI reference in parking access decision-making

2. View System Performance Reports

Admins can access system performance reports that include:

- Daily usage patterns
- Historical data on parking density

This information is very important as a basis for data-driven decision-making and the development of parking systems towards smart campuses or smart cities.



6. Structural Design

After determining the functional needs through the Use Case Diagram, the next step is to design the system structure using the clas diagram. This design defines the key data entities, the attributes they have, and the operational methods that will be run by this AI-based intelligent parking system.

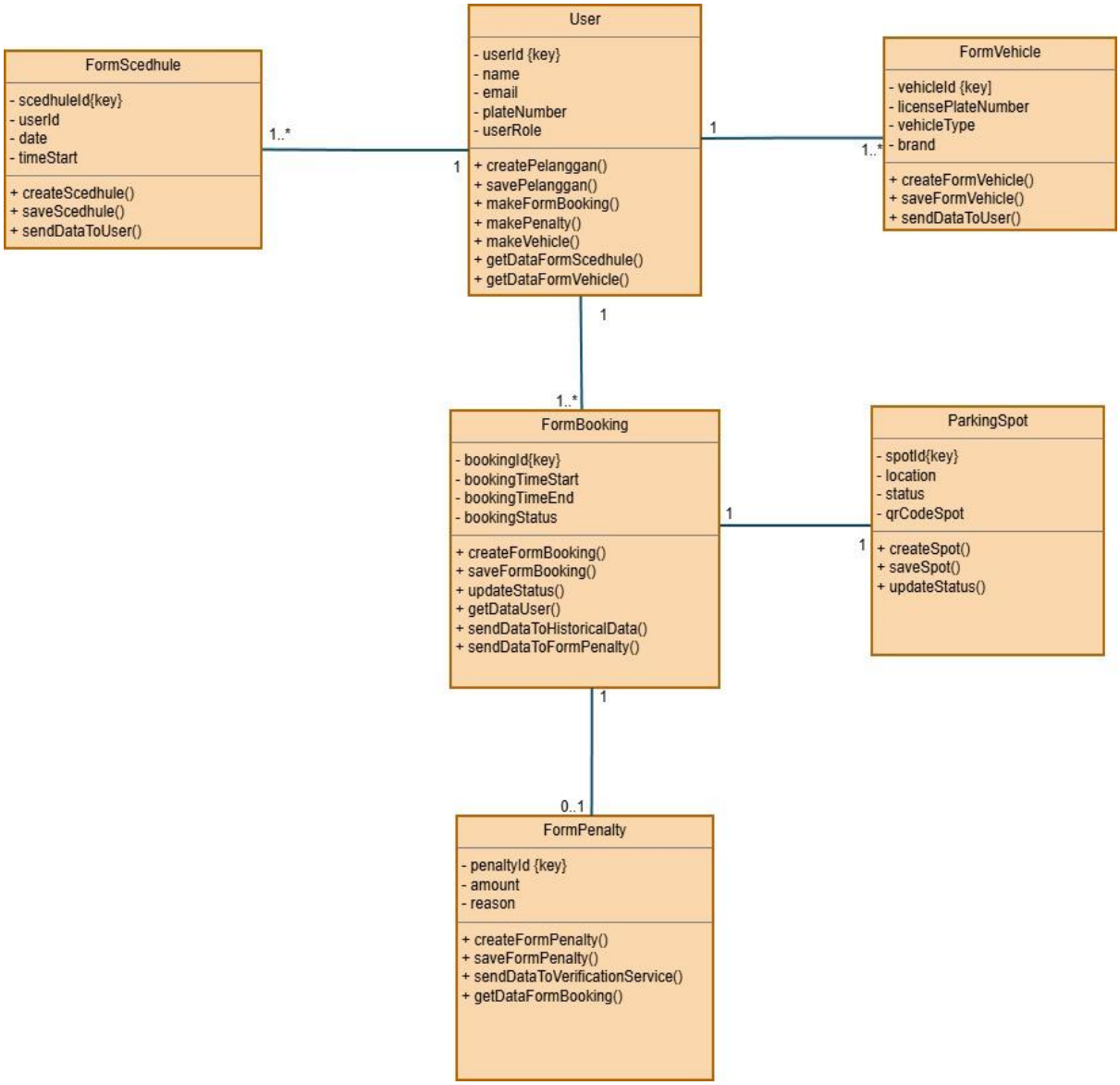
A. Class Component Description

The system consists of six main classes that are integrated with each other to support the reservation process up to automatic verification:

- User: Stores user identities, including roles (userRoles) to distinguish between students, lecturers or staff.
- FormVehicle: Manages the user's vehicle data.
- FormScedhule : Serves as a validation filter. The system will only allow bookings if the user has a college or work schedule on campus on that day.

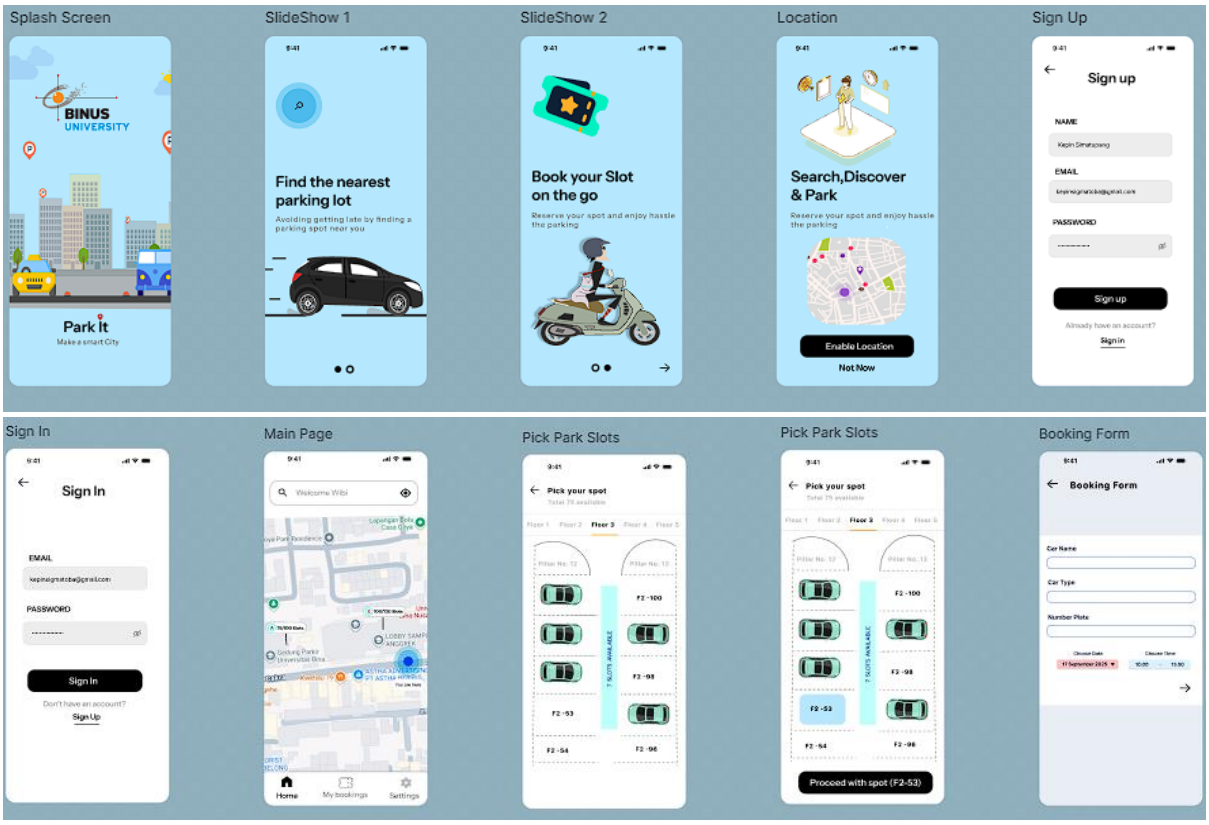
- FormBooking: The heart of the system's transaction that records the user's start time, end time, and reservation status.
- ParkingSpot: Presenting a physical parking slot in the field.
- FormPenalty: An automatic class that records a fine in the event of a violation, such as a non-compliant vehicle or a user who is not present at a specified time.

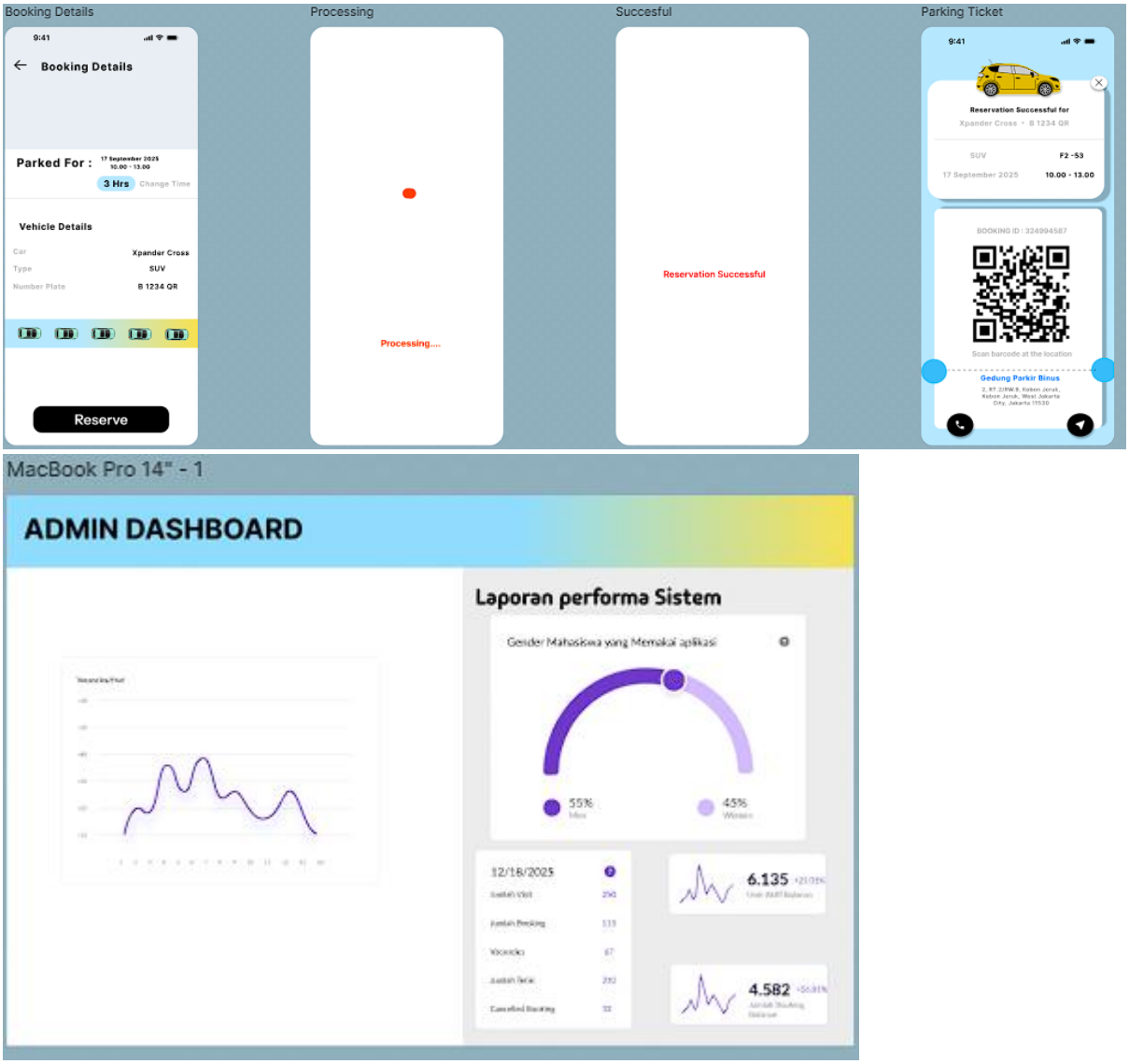
B. Relationship
   1. User to Vehicle & Scedhule (1 – 1..*): One user has control over many vehicle data and schedules, allowing flexibility in the use of facilities.
   2. User to Booking (1 – 1..*) Records the history of parking usage by each individual for historical data needs.
   3. Booking to ParkingSpot (1 – 1): One booking can only occupy one specific parking slot to avoid overlapping reservations.
   4. Booking to Penalty (1 – 0..1): An optional relationship that will only be created if the verification system detects a data discrepancy between the AI-captured license plate number and the data registered in the system



7. User Interface

Tampilan Awal





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